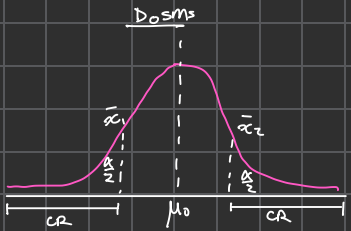


17.1: Interpreting Sample Data using the Normal Distribution:

Assumes the value of the population mean is the one given in the null hypothesis. Then consider the value of the sample mean. If your sample mean is too far away from the assumed population mean, then you conclude that as it is very unlikely that a randomly chosen sample would have such an extreme (relative to H_0) sample mean. Then you can therefore reject the null hypothesis.



Sample Answer:

$H_0: \mu = \mu_0$
$H_1: \mu > \mu_0 \ / \ \mu < \mu_0 \ / \ \mu \neq \mu_0$
} where μ is the population mean of...

State tailedness, $\alpha = \ldots\%$. If two tailed, half α for each tail.

$\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$. Where $\bar{X}$ follows the normal distribution of sample means. If we don't know σ, s can be used only if n is large (as $s \approx \sigma$ almost the same).

Then find $P(\bar{X} \leq \bar{x}) \ / \ P(\bar{X} \geq \bar{x})$, where $\bar{x}$ is the sample mean.
Or find critical regions: $P(\bar{X} \leq \bar{x}_1) \ / \ P(\bar{X} \geq \bar{x}_2)$
Interpret these findings linking to $\alpha / \frac{\alpha}{2}$.

17.2: Bivariate Data: Correlation & Association:

Correlation

If you know both variables are random and you are testing for a linear relationship, test about correlation.
To ensure validity, the underlying population must have a bivariate normal distribution / assume this. This can be seen by a scatter diagram having a roughly elliptical shape.

$H_0: \rho = 0$
$H_1: \rho > 0, \ \rho < 0, \ \rho \neq 0$
} where ρ is the population PMCC value between...

Association

If one, or both, of the variables is not random and/or you are testing for a non linear relationship, you would conduct a test about association. No underlying distribution must be assumed.
Words are used in hypotheses as raw data is not being used, ranks are used.